Guangzhi Xiong, Eric Xie, Amir Hassan Shariatmadari, Sikun Guo, Stefan Bekiranov, and Aidong Zhang. Improving scientific hypothesis generation with knowledge grounded large language models. *arXiv preprint arXiv:2411.02382*, 2024.

Yutaro Yamada, Robert Tjarko Lange, Cong Lu, Shengran Hu, Chris Lu, Jakob Foerster, Jeff Clune, and David Ha. The ai scientist-v2: Workshop-level automated scientific discovery via agentic tree search. *arXiv preprint arXiv:2504.08066*, 2025.

An Yang, Anfeng Li, Baosong Yang, Beichen Zhang, Binyuan Hui, Bo Zheng, Bowen Yu, Chang Gao, Chengen Huang, Chenxu Lv, et al. Qwen3 technical report. *arXiv preprint arXiv:2505.09388*, 2025a.

Yingxuan Yang, Huacan Chai, Yuanyi Song, Siyuan Qi, Muning Wen, Ning Li, Junwei Liao, Haoyi Hu, Jianghao Lin, Gaowei Chang, et al. A survey of ai agent protocols. *arXiv preprint arXiv:2504.16736*, 2025b.

Zijie Yang, Yongjing Yin, Chaojun Kong, Tiange Chi, Wufan Tao, Yue Zhang, and Tian Xu. Shennongalpha: an ai-driven sharing and collaboration platform for intelligent curation, acquisition, and translation of natural medicinal material knowledge. *Cell Discovery*, 11(1):32, 2025c.

Zonglin Yang, Wanhao Liu, Ben Gao, Tong Xie, Yuqiang Li, Wanli Ouyang, Soujanya Poria, Erik Cambria, and Dongzhan Zhou. Moose-chem: Large language models for rediscovering unseen chemistry scientific hypotheses. *arXiv preprint arXiv:2410.07076*, 2024.

Zonglin Yang, Wanhao Liu, Ben Gao, Tong Xie, Yuqiang Li, Wanli Ouyang, Soujanya Poria, Erik Cambria, and Dongzhan Zhou. MOOSE-chem: Large language models for rediscovering unseen chemistry scientific hypotheses. In *The Thirteenth International Conference on Learning Representations*, 2025d. URL https://openreview.net/forum?id=X9OfMNNepI.

Haofei Yu, Zhaochen Hong, Zirui Cheng, Kunlun Zhu, Keyang Xuan, Jinwei Yao, Tao Feng, and Jiaxuan You. Researchtown: Simulator of human research community. *arXiv preprint arXiv:2412.17767*, 2024.

Jiakang Yuan, Xiangchao Yan, Botian Shi, Tao Chen, Wanli Ouyang, Bo Zhang, Lei Bai, Yu Qiao, and Bowen Zhou. Dolphin: Closed-loop open-ended auto-research through thinking, practice, and feedback. *arXiv preprint arXiv:2501.03916*, 2025.

Qiyuan Zhang, Fuyuan Lyu, Zexu Sun, Lei Wang, Weixu Zhang, Wenyue Hua, Haolun Wu, Zhihan Guo, Yufei Wang, Niklas Muennighoff, et al. A survey on test-time scaling in large language models: What, how, where, and how well? *arXiv preprint arXiv:2503.24235*, 2025.

Yusen Zhang, Ruoxi Sun, Yanfei Chen, Tomas Pfister, Rui Zhang, and Sercan Arik. Chain of agents: Large language models collaborating on long-context tasks. *Advances in Neural Information Processing Systems*, 37:132208–132237, 2024.

Yuxiang Zheng, Shichao Sun, Lin Qiu, Dongyu Ru, Cheng Jiayang, Xuefeng Li, Jifan Lin, Binjie Wang, Yun Luo, Renjie Pan, et al. Openresearcher: Unleashing ai for accelerated scientific research. *arXiv preprint arXiv:2408.06941*, 2024.

Minjun Zhu, Yixuan Weng, Linyi Yang, and Yue Zhang. Deepreview: Improving llm-based paper review with human-like deep thinking process. *arXiv preprint arXiv:2503.08569*, 2025.

A. Sampling Time Calculation for Different Types of AI Agents

We referenced existing literature (Guo et al., 2025, Yang et al., 2025a, Muennighoff et al., 2025) and our experience to estimate the sampling time potentially required for different types of AI agents trained via reinforcement learning, as illustrated in Figure 4. Using a hypothetical 671B parameter LLM (similar to Deepseek-R1) running on 8 H100 cards (assuming 40 tokens generated per second), the pure inference time T_{infer_R} for a typical arithmetic reasoning task (generating approximately 10,000 tokens of reasoning content) might be around 250 seconds. For an AI Web Agent, the task might include generating approximately 8,000 tokens of instructions and reports ($T_{infer_WA} \approx 200s$), interspersed with approximately 20 API calls for information search (assuming each search and processing takes $T_{search_API} = 10s$, totaling $T_{search_total} = 20 \times 10s = 200s$), and potentially requiring reading and comprehension of up to 400,000 tokens of web content (assuming reading and comprehension time $T_{read_WA} \approx 200s$). The total sampling time is: $T_{sample_WA} \approx T_{infer_WA} + T_{search_total} + T_{read_WA} \approx 600s$.

In contrast, an AI Scientist executing end-to-end scientific discovery tasks has complexity and interaction requirements far exceeding the previous two types. We roughly estimate it might need to generate over 100,000 tokens of content (for example, operational and experimental code approximately 50,000 tokens ($T_{infer_code} \approx 1250s$), research paper writing approximately 30,000 tokens ($T_{infer_paper} \approx 750s$), reviewing and understanding relevant literature approximately 20,000 tokens ($T_{infer_lit} \approx 500s$)), with pure LLM inference time for just this portion being $T_{infer_CS} = T_{infer_code} + T_{infer_paper} + T_{infer_lit} \approx 2500s$. More critically, the "implementation" process of an AI Scientist, such as code writing, debugging, compiling, running experiments, and data analysis, is highly asynchronous and time-consuming. Assuming a rapid research code operation and experimental cycle (from writing to obtaining preliminary results) requires an average of $T_{op_code} \approx 12$ hours $= 43200s$, while in-depth literature research and analysis might require $T_{op_lit} \approx 20$ minutes $= 1200s$. Therefore, the total estimated sampling time to complete a relatively complete scientific exploration loop would be $T_{sample_CS} = T_{infer_CS} + T_{op_code} + T_{op_lit} \approx 2500s + 43200s + 1200s \approx 46900s$. As intuitively demonstrated in Figure 4, the sampling time required for an AI Scientist (approximately 46,000 seconds) far exceeds that of an AI Reasoner (approximately 250 seconds) and an AI Web Agent (approximately 700 seconds). Notably, AI Reasoners can typically rapidly generate large quantities of training samples through batch generation in parallel, whereas each implementation step of an AI Scientist (especially parts involving code execution and experiment waiting) is almost entirely asynchronous, and requires exclusive computational resources or experimental equipment for learning and feedback collection during operations. Consequently, in actual reinforcement learning training processes, the disparity in real training duration between AI Scientists and the former two types will be even more pronounced.

For the calculation of human duration, we referenced existing metrics. For instance, for reasoning tasks, we referred to the human time from the International Mathematical Olympiad, which is approximately 1.5 hours per problem. For Web Agent tasks, we adopted the average human problem-solving time from BrowseComp (Wei et al., 2025) (2 hours) as the human standard. For Scientist tasks, although each paper often requires months of collaborative work by multiple people, for ease of calculation, we used the human duration of 48 hours from PaperBench (Starace et al., 2025) for statistics; however, even under these conditions, humans achieve a success rate of less than 50%.

Table 4: Timeline of AI Scientist Ideas and Code Implementations by Month

	2024-08	2024-09	2024-10	2024-11	2024-12	2025-01	2025-02	2025-03	2025-04
w/o Exp	(Zheng et al., 2024)	(Ghafarollahi and Buehler, 2024), (Radensky et al., 2024)	(Pu et al., 2025a), (Yang et al., 2024), (Su et al., 2024), (Li et al., 2024a), (Hu et al., 2024), (Liu et al., 2025), (Wang et al., 2024b)	(Weng et al., 2025), (Xiong et al., 2024)	(Gu et al., 2024a), (Li et al., 2024b), (Yu et al., 2024)		(Gottweis et al., 2025)	(Rabby et al., 2025), (Saeedi et al., 2025)	(O'Neill et al., 2025), (Garikaparthi et al., 2025), (Sanyal et al., 2025)
w/ Exp	(Lu et al., 2024), (Li et al., 2024c)			(Liu et al., 2024b), (Liu et al., 2024a)		(Yuan et al., 2025), (Schmidgall et al., 2025)	(Jiang et al., 2025), (Kon et al., 2025)	(Schmidgall et al., 2025), (Jansen et al., 2025)	(Yamada et al., 2025), (Seo et al., 2025)

B. Regarding the statistics for the papers

We have conducted a comprehensive search on arXiv to gather relevant publications in the AI Scientist field. This collection includes a series of papers from August 2024 to April 2025 for methods or systems, which are cited in Table 4. It indicates that, to date, a significant number of papers have focused on Idea Generation tasks, often without concrete implementations.

Nevertheless, an encouraging trend has emerged since early 2025. As illustrated in Figure 3, implementation-focused research has demonstrated stronger growth momentum, with incremental growth nearly matching that of non-implementation studies by Spring 2025. This suggests the community is beginning to recognize the critical importance of implementation capabilities for developing truly effective AI Scientists—moving beyond theoretical constructs toward practical systems capable of reliable execution.